Brahma Vamsi Madasu
Senior Data Engineer

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Professional summary:

- With 11+ years of extensive IT experience, specializing in Azure Cloud Services, Big Data technologies, data modelling, ETL/ELT development, validation, deployment, monitoring, visualization, and requirement gathering across healthcare and finance domains.
- Demonstrated expertise in **Azure Cloud Platform**: Azure Data Factory (ADF), Azure Databricks (PySpark, Spark SQL), Azure Synapse Analytics, Azure Data Lake Storage Gen2, Blob Storage, Azure SQL Database, Event Hubs, Logic Apps, Functions, Purview, Key Vault, DevOps, Repos, Monitor, Cosmos DB, AKS, and HDInsight.
- Skilled in **ETL/ELT pipeline design and orchestration** using **Azure Data Factory and Databricks**, implementing incremental loads, CDC, surrogate key strategies, and complex dataflows to process terabytes of structured and semi-structured data.
- Hands-on expertise in **migrating legacy ETL workloads** (SSIS, Informatica, DataStage, Teradata utilities, Oracle PL/SQL) into **Azure Synapse Analytics, Databricks, and Snowflake**, improving performance by 30–40% and reducing operational costs.
- Architected and deployed **Delta Lake Medallion Architecture (Bronze–Silver–Gold layers)** with schema enforcement, ACID transactions, and lineage tracking to manage healthcare PHI data and financial datasets.
- Strong background in **healthcare data engineering**: integrated **Electronic Health Records (Epic, Cerner), payer claims (EDI 837/835), HL7/FHIR feeds, IoT patient monitoring devices, lab systems**, and CRM data into enterprise data lakes and warehouses.
- Designed **healthcare-specific warehouse models** (fact: claims, encounters, pharmacy, authorizations; dim: member, provider, diagnosis, plan) to support payer analytics, utilization management, CMS/HEDIS reporting, and quality improvement initiatives.
- Finance sector expertise: developed **data warehouse schemas for transactions, accounts, payments, products, customers, and channels**; supported fraud detection, AML, compliance, and regulatory reporting.
- Implemented **real-time streaming frameworks** using Kafka, Event Hubs, Spark Streaming, and Azure Stream Analytics to process **HL7 ADT events, IoT data, and financial transactions** with low latency for alerts and fraud/risk scoring.
- Built and optimized **PySpark/Spark SQL transformations** for cleansing, deduplication, and aggregation across SQL Server, Oracle, PostgreSQL, Teradata, MongoDB, and MySQL.
- Developed **Snowflake data marts** using surrogate keys, clustering keys, materialized views, Slowly Changing Dimensions (SCD), and zero-copy cloning, achieving 25% improvement in query performance.
- Implemented **DBT models** for reusable healthcare transformations, automated testing, documentation, and HIPAA compliance audits.
- Designed Power BI and Tableau dashboards for executives and clinicians, visualizing **readmission rates, LOS, chronic disease stratification, provider performance, fraud detection, portfolio risk, and cost/utilization trends**.
- Configured **Azure Monitor, Purview, and Logic Apps** to provide real-time pipeline monitoring, metadata management, lineage tracking, and automated alerting, cutting incident response times by 30%.
- Designed and implemented **MLOps pipelines in Databricks with MLflow and Azure ML**, deploying predictive models for **readmission risk, chronic disease progression, and fraud detection** in production.
- Led **data governance and compliance programs**: applied Azure RBAC, Key Vault, tokenization, and encryption at rest/in transit to secure PHI/PII and achieve HIPAA/HITECH, PCI-DSS, and SOX compliance.
- Automated infrastructure provisioning with **Terraform and Azure DevOps Pipelines**, deploying ADF, Databricks, Synapse, Event Hubs, Snowflake, and secure VNETs across dev, QA, and prod.
- Migrated **SSIS, Informatica, and DataStage ETL** workflows into **ADF pipelines and Databricks notebooks**, modernizing pipelines with reusable templates and error handling.
- Extensive **Big Data background**: Hadoop (HDFS, YARN, MapReduce), Hive, Pig, Sqoop, Oozie, NiFi, Spark (RDD, Data Frames, Streaming), HBase, MongoDB, and AWS EMR/Redshift.
- Developed **Spark Streaming apps** using Scala, Kafka, and HBase for real-time analytics and Lambda architecture pipelines.
- Designed Hive schemas with partitioning, bucketing, ORC/Parquet formats, and Snappy compression; optimized HiveQL queries for performance and scalability.

- Skilled in **query optimization and Spark tuning**, using broadcast joins, caching, and partitioning to cut processing times significantly.
- Designed and modelled normalized, denormalized, relational, dimensional, Star and Snowflake schemas using Erwin and ER/Studio, supporting both OLTP and OLAP applications.
- Proficient in **Python, PySpark, SQL (T-SQL, PL/SQL, HiveQL), Scala, Java**, with strong shell scripting (Linux/Unix, PowerShell) for automation, data manipulation, and deployment.
- Delivered **data validation, profiling, and reconciliation frameworks** to ensure high data quality across healthcare and finance reporting pipelines.
- Experienced with **CI/CD automation and DevOps**: Git, GitHub, GitLab, Jenkins, GitHub Actions, Azure DevOps YAML pipelines, Docker, Kubernetes, Helm, Terraform.
- Experienced in **Agile/Scrum methodology**, participating in sprint planning, daily stand-ups, backlog grooming, retrospectives, and mentoring junior engineers in Azure and Big Data best practices.

Education:

- Master's in data science, University of New Haven, CT **Jan 2012 to Dec 2013**
- Bachelor's in Electronics and communication Engineering, Sathyabama University **Jun 2007 to Jun 2011**

Certifications:

- Microsoft Certified Azure Data Engineer Associate – DP 203
- Microsoft Certified Fabric Data Engineer Associate -DP 700
- Microsoft Azure Fundamentals AZ-900
- Databricks Certified Data Engineer Professional

Technical skills:

Azure Services	Azure Data Factory (ADF), Azure Databricks, Azure Synapse Analytics, Azure Data Lake Storage Gen2 (ADLS), Azure Blob Storage, Azure SQL Database, Azure Event Hubs, Azure Functions, Azure Logic Apps, Azure Purview, Azure Key Vault, Azure DevOps, Azure Repos, Azure Monitor, Azure Cosmos DB, Azure Kubernetes Service (AKS), Azure Stream Analytics, Azure Virtual Machines, Azure Active Directory, Azure HDInsight.
AWS Services	Amazon S3, AWS Glue, AWS Redshift, AWS EMR, AWS Lambda, AWS Step Functions, Amazon Kinesis, AWS Athena, AWS Lake Formation, AWS CloudWatch, AWS EventBridge, AWS Secrets Manager.
Big Data Technologies	Hadoop (HDFS, MapReduce, YARN), Apache Spark (PySpark, Spark SQL, Spark Streaming), Hive, Pig, Sqoop, Kafka, Oozie, Airflow, Delta Lake, DBT, Snowflake, Informatica, DataStage, Talend, Apache NiFi, Cloudera, Hortonworks, Zookeeper, Elasticsearch, Kibana.
Languages	SQL, T-SQL, PL/SQL, PySpark, Python (Pandas, NumPy), Scala, Java, HiveQL, APIs (REST/SOAP).
Web Technologies	HTML, CSS, JavaScript, XML, JSP, Restful, SOAP
Operating Systems	Windows (10/7/XP/2000/NT/98/95) UNIX, LINUX, UBUNTU, OS.
File formats	ORC, Avro, CSV, JSON, TXT, XML, Excel
Build Automation tools	Maven, SBT
Version Control & CI/CD Tools	Git, GitHub, Bitbucket, GitLab, Jenkins, GitHub Actions, Azure DevOps Pipelines (YAML), Terraform, Docker, Kubernetes, Helm, RBAC/IAM Policies, Bicep, CloudFormation.
IDE &Build Tools, Design	Eclipse, Visual Studio, SSIS, SSAS, SSRS, SSMS
Visualization Tools	Power BI, Tableau, Microsoft Fabric, QlikView, Alteryx.
Databases	Azure Synapse, Snowflake, SQL Server, Oracle, PostgreSQL, MySQL, MongoDB, Teradata, DynamoDB, Cassandra.

Professional experience:

Client: Evolent, Tx. **Sep 2022 - Present**
Role: Senior Data Engineer
Responsibilities:

- Designed and implemented **enterprise-scale healthcare data pipelines** using Azure Data Factory (ADF) and Databricks to ingest and transform **EHR (Epic, Cerner), payer claims (EDI 837/835), HL7/FHIR feeds, IoT medical devices, and lab systems**, ensuring seamless integration across multiple data sources.
- Migrated large-scale **on-premise Oracle and Teradata ETL processes** to Azure Synapse Analytics and Snowflake, reducing processing times by 40% and eliminating legacy system dependencies.
- Architected and deployed **Delta Lake Medallion Architecture (Bronze–Silver–Gold layers)** in ADLS Gen2, supporting **ACID compliance, schema enforcement, and lineage tracking** for PHI across healthcare domains such as claims, encounters, member, provider, and pharmacy data.
- Built **real-time data streaming frameworks** with Kafka, Azure Event Hubs, and Spark Streaming to process HL7 ADT messages and IoT-generated patient monitoring data, providing clinicians and care managers with near real-time alerts.
- Developed advanced **PySpark and Spark SQL transformations** to clean, normalize, and aggregate healthcare datasets from SQL Server, PostgreSQL, Oracle, MongoDB, and MySQL, enabling unified patient and claims analytics.
- Designed **healthcare-specific data warehouse models** (fact: claims, encounters, pharmacy, authorizations; dim: member, provider, diagnosis, procedure, plan) to support payer and provider analytics, **risk adjustment**, **utilization management**, and **quality reporting (HEDIS, CMS Star Ratings)**.
- Optimized **Snowflake and Synapse** workloads by implementing clustering keys, materialized views, partitioning strategies, and zero-copy cloning, improving query performance by 25% and reducing storage costs.
- Developed **incremental load strategies** in ADF using surrogate keys, lookup activities, and CDC techniques to process terabytes of claims and patient data with minimal downtime.
- Automated **data ingestion and scheduling** using Airflow DAGs, ADF pipelines, and Logic Apps, ensuring timely delivery of payer and provider reports across multiple healthcare systems.
- Implemented **DBT for healthcare data transformations**, enabling reusable data models, automated data quality checks, metadata documentation, and faster compliance audits.
- Designed and deployed **Power BI dashboards and Tableau reports** for clinical and financial stakeholders, providing interactive visualizations for readmission rates, length of stay (LOS), chronic disease risk stratification, provider performance, and cost/utilization trends.
- Configured **Azure Monitor and Logic Apps** to create real-time monitoring and alerting for healthcare data pipelines, reducing incident response times by 35%.
- Led the design and implementation of **MLOps pipelines** in Databricks using MLflow and Azure ML, deploying predictive models for patient readmission risk, chronic disease progression, and care gap detection.
- Partnered with clinicians, payer IT teams, and compliance officers to integrate **Enterprise Risk Management (ERM) strategies**, enhancing HIPAA/HITECH compliance, audit readiness, and regulatory reporting accuracy.
- Implemented **robust data governance frameworks** using Azure Purview, Azure RBAC, Azure Key Vault, and Active Directory groups to ensure secure, role-based access to PHI across dev, QA, and production environments.
- Automated infrastructure provisioning using **Terraform** to deploy HIPAA-compliant Azure resources including ADF, Databricks, Synapse, Event Hubs, and secure VNETs, reducing manual configuration errors.
- Migrated legacy **SSIS and DataStage ETL workflows** into ADF and Databricks notebooks, optimizing performance and modernizing integration with cloud-native services.
- Created **Python and Scala scripts in Databricks** for validation, cleansing, deduplication, and transformation of healthcare data, improving data quality across downstream analytics systems.
- Designed **centralized access governance models** to enable secure cross-team collaboration for payer and provider data analysis while meeting compliance requirements.
- Applied **data masking, tokenization, and encryption at rest/in transit** strategies for PHI using Key Vault, ensuring secure workflows for healthcare data processing.
- Built **batch and streaming data solutions** combining Kafka, Spark, and Event Hubs with Azure Analysis Services for **real-time insights into claims adjudication, provider performance, and patient monitoring**.
- Configured **Databricks Unity Catalog and Purview integration** to streamline healthcare data cataloguing, metadata management, and lineage tracking.
- Implemented **Agile development practices**, participating in sprint planning, backlog grooming, and daily stand-ups while mentoring offshore and junior engineers on Azure data engineering and healthcare compliance best practices.
- Conducted **code reviews, performance tuning, and pipeline optimization**, ensuring all healthcare data engineering solutions adhered to compliance, scalability, and operational SLAs.

- Collaborated with business stakeholders to design **executive dashboards** for cost-of-care, value-based care programs, and population health management initiatives, directly impacting organizational strategy.
- Led **cross-functional initiatives** with clinical operations, IT, and compliance teams to align healthcare data engineering solutions with enterprise goals, reducing manual reporting efforts by 60%.

Environment: Azure Data Factory, Azure Databricks, PySpark, Spark SQL, Delta Lake, ADLS Gen2, Azure Synapse, Snowflake, Azure SQL DB, Event Hubs, Kafka, Spark Streaming, HDInsight, PostgreSQL, Oracle, Teradata, MongoDB, DBT, Airflow, Terraform, Azure DevOps, GitHub, Jenkins, Power BI, Tableau, Azure Purview, Azure Key Vault, HIPAA, HITECH, HL7, FHIR, EDI 837/835, IoT healthcare devices.

Client: Discover, IL. **July 2019 to Aug 2022**
Role: Azure data engineer
Responsibilities:

- Designed and implemented **Azure Data Factory (ADF) pipelines** to ingest, transform, and load **transactional banking, customer, and credit data** from diverse on-premises systems (SQL Server, Oracle, PostgreSQL, Teradata) into **Azure Synapse Analytics and ADLS Gen2**, ensuring high availability and data quality
- Migrated large-scale **on-premise ETL workloads** (SSIS, Informatica, Teradata utilities) into **Azure Databricks and Synapse**, reducing batch processing times by 35% and optimizing operational costs.
- Developed **end-to-end ETL processes** with PySpark in Databricks to process terabytes of financial data, including **customer transactions, deposits, credit card usage, loan servicing, and payments**, ensuring accuracy for reporting and regulatory compliance.
- Built **data Lakehouse solutions** on ADLS Gen2 and Delta Lake, supporting both batch and real-time ingestion pipelines for transactional and customer data.
- Designed **data warehouse schemas** (fact tables: transactions, accounts, payments; dimension tables: customer, branch, product, channel) to support **financial reporting, customer insights, and risk management analytics**.
- Implemented **real-time streaming solutions** using Kafka, Event Hubs, and Spark Streaming to capture banking transactions and payment events, supporting **fraud detection, AML (Anti-Money Laundering), and risk scoring** with minimal latency.
- Developed **Snowflake data marts** with surrogate keys, Slowly Changing Dimensions (SCD), clustering keys, and materialized views to optimize compliance reporting and reconciliation workflows.
- Implemented **incremental replication strategies** in ADF with CDC (Change Data Capture), ensuring accurate daily loads of high-volume transactional data.
- Automated **data ingestion and orchestration** using Airflow DAGs and ADF triggers, ensuring SLA compliance for risk and compliance reporting deadlines.
- Created **SQL stored procedures, views, and functions** for reconciliation of financial datasets across multiple systems, enabling internal audit and external regulatory reporting.
- Partnered with compliance and risk teams to align pipelines with **PCI-DSS, SOX, and Basel III** standards, ensuring end-to-end governance and audit readiness.
- Secured sensitive **PCI/PII data** by configuring Azure RBAC, Azure Key Vault, and encryption at rest/in transit, reducing security risks and supporting regulatory compliance.
- Designed **Power BI dashboards** to visualize KPIs such as **transaction volumes, fraud detection metrics, customer churn, loan delinquency, and portfolio performance**, enabling executives to make data-driven decisions.
- Implemented **CI/CD pipelines** with Azure DevOps and GitHub Actions for automated deployment of ETL pipelines, reducing manual errors and accelerating release cycles.
- Optimized **Databricks Spark jobs** using partitioning, caching, and broadcast joins to improve runtime performance for large transaction datasets.
- Automated infrastructure provisioning using **Terraform**, creating repeatable deployments of Databricks clusters, Synapse pools, and secure VNET configurations across dev, QA, and prod.
- Migrated **legacy SSIS ETL packages** into ADF, standardizing pipelines and integrating with Azure monitoring for better reliability.
- Collaborated with QA and finance teams to validate ETL outputs against **reconciliation rules, balance checks, and compliance policies**, ensuring accuracy in financial reporting.
- Worked in **Agile/Scrum environments**, participating in sprint planning, backlog grooming, and daily stand-ups, while mentoring junior engineers on Azure and big data best practices.

Environment: Azure Data Factory, Azure Databricks, PySpark, Spark SQL, Delta Lake, ADLS Gen2, Azure Synapse, Snowflake, SQL Server, Oracle, PostgreSQL, Teradata, Kafka, Event Hubs, Spark Streaming, SSIS, Informatica, Airflow, Terraform, Azure Key Vault,

Azure DevOps, GitHub, Jenkins, Power BI, Tableau, PCI-DSS, SOX, Basel III compliance.

Client: AXA insurance, CT. **May 2016 to June2019**
Role: Big Data Developer
Responsibilities:

- Developed **NiFi workflows** to automate secure data movement between heterogeneous Hadoop systems, ensuring consistent ingestion and transformation pipelines.
- Configured, deployed, and maintained **multi-node Hadoop/YARN clusters** and **Kinesis Streams** for real-time financial and operational data processing.
- Built **Spark applications (RDD, Data Frames, Spark SQL)** over Cloudera Hadoop for analytics on structured and unstructured datasets, improving batch ETL efficiency compared to MapReduce.
- Developed **Scala and Java MapReduce programs** to process large volumes of log, transaction, and customer data, implementing aggregation, filtering, and anomaly detection logic.
- Designed **Hive schemas with partitioning, bucketing, and Snappy compression**, and converted data into **Parquet** formats for optimized querying and storage.
- Imported **large datasets from DB2 and RDBMS** into Hive using **Sqoop**; created automated Sqoop jobs for incremental data ingestion into Hadoop clusters.
- Implemented **Apache Pig scripts** for data cleansing, transformation, and enrichment; optimized Pig UDFs in Java/Scala to handle complex ETL requirements.
- Coordinated **ETL pipelines** across Hive, Pig, and Spark jobs using **Oozie workflows and coordinators**, ensuring timely availability of curated datasets for downstream analytics.
- Designed and executed **Spark Streaming applications with Lambda Architecture**, processing real-time data feeds from DB2 into HBase tables via Kinesis Streams.
- Integrated **MongoDB** with Spark for interactive queries and processed semi-structured JSON datasets for analytical workloads.
- Built **HBase tables** for fast lookups and random access to large healthcare/financial datasets, supporting operational dashboards and APIs.
- Used **Spark SQL and HiveQL** for interactive ad-hoc queries, supporting business analysts with reporting and predictive modeling data sets.
- Performed **data extraction, profiling, and transformation** across diverse sources (RDBMS, APIs, log files) into Hadoop Data Lake, enabling advanced analytics.
- Wrote **Pig and Hive scripts** to clean, enrich, and partition daily transaction and clickstream data, supporting customer segmentation and churn analysis.
- Developed **Spark jobs for in-memory computation**, enhancing performance for iterative machine learning algorithms and complex transformations.
- Worked with **AWS Cloud services** (EC2, S3, EBS, RDS, VPC) to integrate Hadoop/Spark workloads with cloud-native storage and compute.
- Analyzed **SQL scripts and legacy ETL logic** and re-engineered them into Spark pipelines, reducing batch runtime by up to 40%.
- Migrated existing **Hive/SQL queries** into Spark SQL and Scala-based transformations, improving query performance and scalability.
- Developed **custom Spark UDFs** and aggregation functions to extend query capabilities for financial and customer behavior datasets.
- Responsible for **Kinesis → Spark → HBase integration** pipelines to deliver real-time curated datasets to customer-facing APIs.
- Built **data ingestion pipelines with Flume and Kafka** for streaming log and transactional data, later transformed into Hive tables for reporting.
- Provided **support for BI teams**, designing Hive external tables, creating data marts, and validating ETL outputs for reporting.
- Mentored junior developers in Hadoop/Spark best practices and contributed to Agile sprint planning and daily stand-ups., Kafka, Flume, AWS (EC2, S3, RDS, VPC), Cloudera, Hortonworks, Yarn, Linux.

Environment: Hadoop, HDFS, Hive, Pig, Sqoop, Spark (RDD, SQL, Streaming, Data Frames), Scala, Java, Python, Shell Scripting, MapReduce, HBase, Kafka, Flume, Oozie, AWS (EC2, S3, RDS, VPC), MongoDB, Cloudera Manager, Hortonworks, Yarn, Ubuntu/Linux, Agile.

Client: Hilmar Company, CA. **Jan 2014 to Apr 2016**
Role: SQL developer
Responsibilities:

- Designed, developed, and optimized **complex SQL queries, stored procedures, triggers, views, and user-defined functions** in SQL Server, Oracle PL/SQL, and MySQL to support enterprise reporting and operational analytics.
- Built and maintained **SSIS ETL workflows** to extract, transform, and load financial, operational, and healthcare data from multiple OLTP systems into staging and reporting databases, ensuring accuracy, lineage, and completeness.
- Modelled and implemented **relational, dimensional, star, and snowflake schemas**, supporting both OLTP applications and OLAP/BI use cases for executives and operational teams.
- Performed **query optimization and indexing strategies** (clustered/non-clustered indexes, execution plan analysis) that reduced execution times for mission-critical financial reports by up to 50%.
- Developed **parameterized stored procedures** for incremental loads, reconciliation, audit logging, and data quality checks, automating recurring ETL workflows and reducing manual intervention.
- Migrated legacy **MS Access and flat-file pipelines** into SQL Server and Oracle, eliminating manual reporting inefficiencies and standardizing ETL processes.
- Built **error handling and logging frameworks** in SSIS and PL/SQL to provide robust monitoring and recovery for daily ETL jobs.
- Developed automation scripts in **T-SQL and PL/SQL** for recurring data loads, reconciliation checks, and financial audits, ensuring compliance with internal and regulatory policies.
- Supported **financial and healthcare units** by delivering ad-hoc SQL queries and reports for compliance audits, operational KPIs, and strategic planning.
- Integrated **SQL Server with SSRS and Tableau** dashboards to deliver interactive reports on transactions, claims, and operational metrics for executives.
- Provided **database administration support** (user access, backup/restore, indexing, performance monitoring) across development, QA, and production environments.
- Participated in **full SDLC lifecycle** including requirements gathering, database design, ETL development, testing, deployment, and ongoing production support.
- Collaborated in **Agile teams**, participating in sprint ceremonies and mentoring junior developers on SQL optimization and ETL best practices.

Environment: SQL Server, T-SQL, SSIS, SSRS, Oracle PL/SQL, MySQL, ETL, Stored Procedures, Functions, Views, Indexing, Query Optimization, Data Warehousing, Star/Snowflake Schemas, Performance Tuning, Tableau, Agile.